Continuity and the Intermediate Value Theorem

ANSWER KEY

Section 1: Three-Part Definition of Continuity

1 Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ **is defined,** $\lim_{x \to c} f(x)$ **exists, and** $\lim_{x \to c} f(x) = f(c)$ ✓

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $f(c)$ is continuous on (a, b), $f(c) = 0$, and $f'(c) \neq 0$

d. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

By the formal definition of continuity at a point, $f(c)$ must be defined, the two-sided limit must exist, and the limit must equal the function value.

2 Let $f(x) = \frac{x^2 - 9}{x - 3}$ for $x \neq 3$. What value must be assigned to $f(3)$ to make f continuous at $x = 3$?

a. 3

b. 6 ✓

c. 0

d. No value can make it continuous

The limit is $\lim_{x \to 3} \frac{(x - 3)(x + 3)}{x - 3} = 3 + 3 = 6$. Therefore, $f(3)$ must equal 6.

3 The function $g(x) = \frac{1}{x - 2}$ has what type of discontinuity at $x = 2$?

a. **Infinite (essential) discontinuity** ✓

b. Removable discontinuity

c. Jump discontinuity

d. Oscillating discontinuity

Because $\lim_{x \to 2^+} g(x) = \infty$ and $\lim_{x \to 2^-} g(x) = -\infty$, there is an infinite discontinuity (vertical asymptote) at $x = 2$.

Section 2: Piecewise Functions & Continuity

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4 For what value of k is $f(x) = kx^2 + 1$ for $x \leq 2$, and $f(x) = 3x - 1$ for $x > 2$ continuous at $x = 2$?

a. 1 ✓

b. 2

c. 3

d. 0

Equate left and right limits at $x = 2$: $k(2^2) + 1 = 3(2) - 1 \Rightarrow 4k + 1 = 5 \Rightarrow 4k = 4 \Rightarrow k = 1$.

5 If $g(x) = \frac{\sin(ax)}{x}$ for $x \neq 0$, and $g(0) = 4$, what value of a ensures continuity at $x = 0$?

a. 1

b. 2

c. 4 ✓

d. $\frac{1}{4}$

$\lim_{x \to 0} \frac{\sin(ax)}{x} = a$. For continuity, $a = g(0) = 4$.

6 The function $h(x) = x + 2$ for $x < 1$, and $h(x) = 2x + 1$ for $x \geq 1$ has what property at $x = 1$?

a. **Continuous because both one-sided limits equal** 3 ✓

b. Jump discontinuity of size 1

c. Infinite discontinuity

d. Undefined at $x = 1$

$\lim_{x \to 1^-} (x + 2) = 3$ and $\lim_{x \to 1^+} (2x + 1) = 3 = h(1)$. Therefore, h is continuous at $x = 1$.

Section 3: Intermediate Value Theorem (IVT)

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7 What are the hypotheses of the Intermediate Value Theorem for a function f on interval $[a, b]$?

a. f **must be continuous on the closed interval** $[a, b]$ ✓

b. f must be differentiable on (a, b)

c. $f(a)$ and $f(b)$ must have opposite signs

d. f must be strictly increasing on $[a, b]$

The Intermediate Value Theorem requires only that f is continuous on the closed interval $[a, b]$.

8 Let f be continuous on $[1, 5]$ with $f(1) = -3$ and $f(5) = 7$. Does the IVT guarantee a solution to $f(c) = 0$ for some $c \in (1, 5)$?

a. **Yes, because** 0 **lies between** -3 **and** 7 ✓

b. No, f must also be differentiable

c. No, f must be a polynomial

d. Only if $f'(x) > 0$ for all x

Because f is continuous on $[1, 5]$ and $f(1) < 0 < f(5)$, the IVT guarantees at least one $c \in (1, 5)$ such that $f(c) = 0$.

9 On which interval is the polynomial $P(x) = x^3 - 3x - 1$ guaranteed to have a real zero by the IVT?

a. $[1, 2]$ ✓

b. $[0, 1]$

c. $[-1, 0]$

d. $[2, 3]$

$P(1) = 1 - 3 - 1 = -3 < 0$, and $P(2) = 8 - 6 - 1 = 1 > 0$. By IVT, a zero exists in $(1, 2)$.

Section 4: Extreme Value Theorem (EVT)

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10 What does the Extreme Value Theorem guarantee for a function f continuous on a closed interval $[a, b]$?

a. f **attains both an absolute maximum and an absolute minimum on** $[a, b]$ ✓

b. f has a horizontal tangent on (a, b)

c. $f'(c) = 0$ for some $c \in (a, b)$

d. f is invertible on $[a, b]$

The EVT states that any continuous function on a closed bounded interval $[a, b]$ must attain an absolute maximum and an absolute minimum.

11 Why does the function $f(x) = \frac{1}{x}$ NOT attain an absolute maximum on $(0, 2]$?

a. **The interval is open at** 0 **and** f **increases without bound as** $x \to 0^+$ ✓

b. f is not differentiable on $(0, 2]$

c. f is bounded on $(0, 2]$

d. $f(2) = 0.5$ is the maximum

The interval $(0, 2]$ is not closed, so the EVT does not apply. As $x \to 0^+$, $f(x) \to \infty$, so no absolute maximum exists.

12 Find the absolute maximum value of $f(x) = -x^2 + 4x + 1$ on the closed interval $[0, 3]$.

a. 5 ✓

b. 4

c. 1

d. 6

Critical point: $f'(x) = -2x + 4 = 0 \Rightarrow x = 2$. Evaluate:
$f(0) = 1, f(2) = -(4) + 8 + 1 = 5, f(3) = -(9) + 12 + 1 = 4$. Absolute maximum is 5.

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